MINI PROJECT REVIEW 1

SCHEMA DIAGRAM, ER DIAGRAM AND DDL COMMANDS

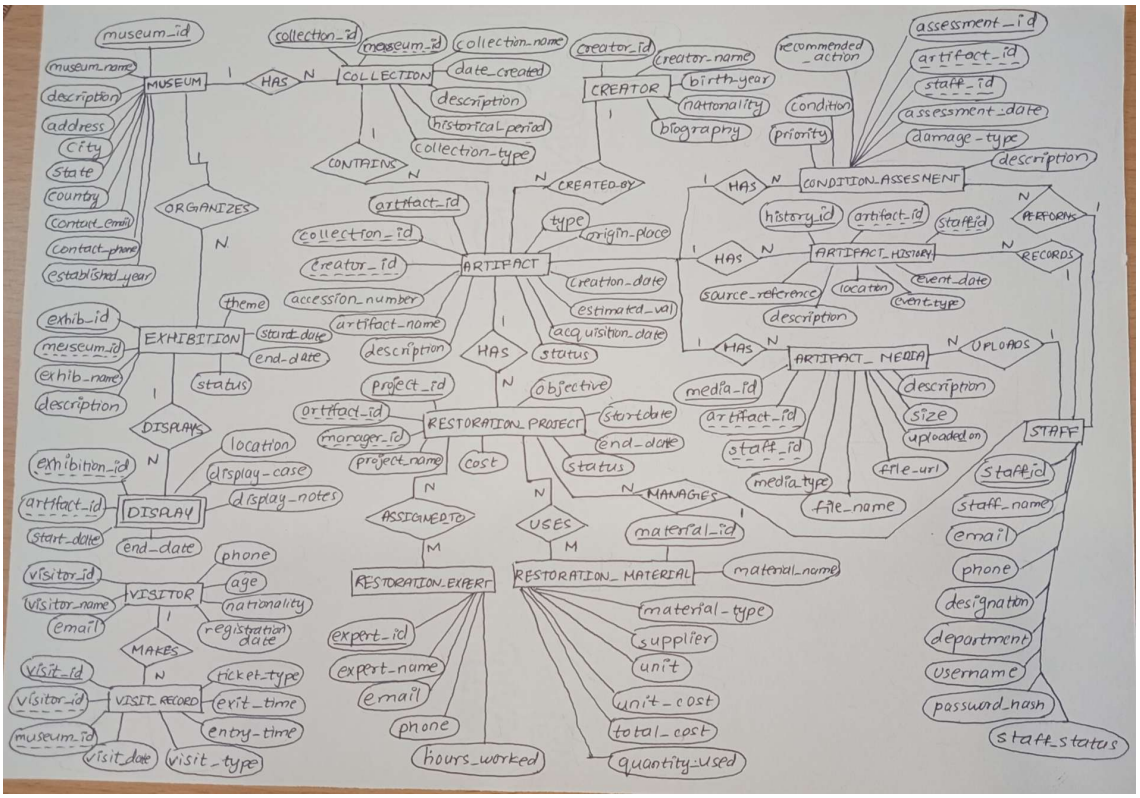
Name: Riona Jeena Dsouza

SRN: PES2UG24CS654

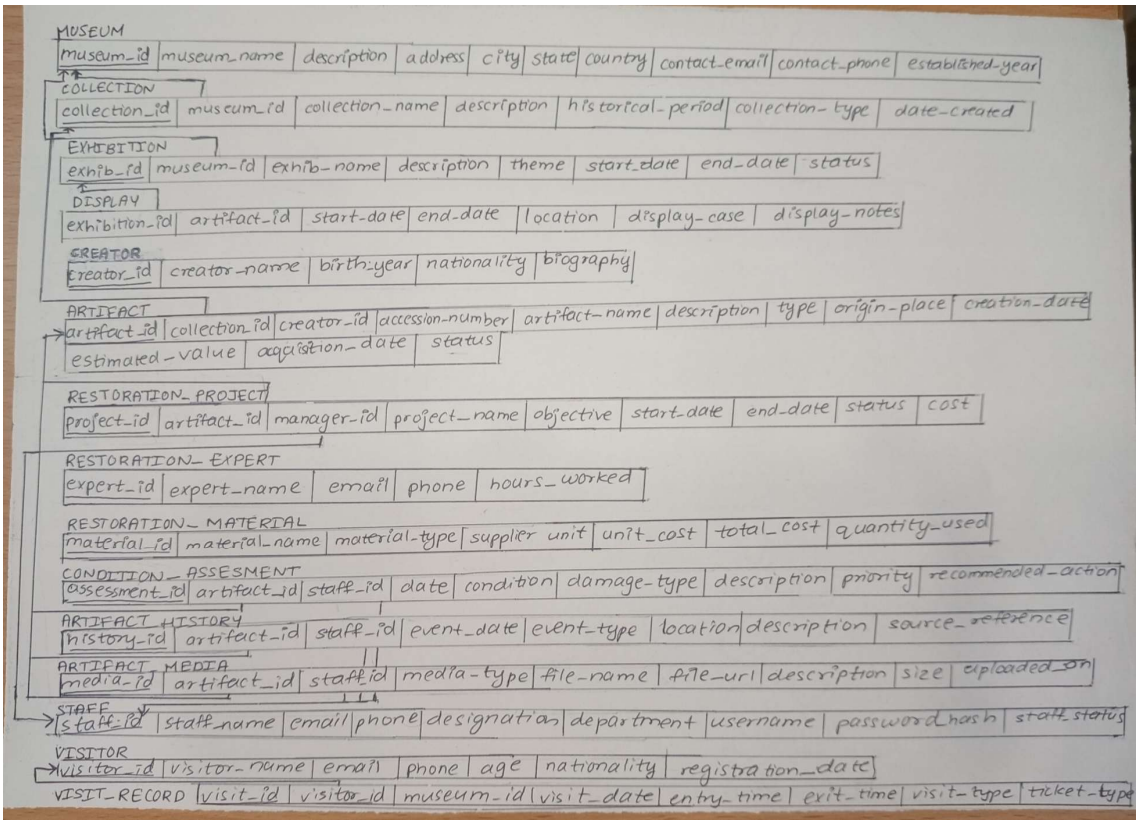
Name: Ramya

SRN: PES2UG24CS653

ER Diagram:



Schema Diagram:



DDL Statements:

```
mysql> CREATE DATABASE digital_museum_db;
Query OK, 1 row affected (0.05 sec)

mysql> USE digital_museum_db;
Database changed
```

```
mysql> CREATE TABLE MUSEUM (
    ->     museum_id INT AUTO_INCREMENT,
    ->     museum_name VARCHAR(150) NOT NULL,
    ->     description TEXT,
    ->     address VARCHAR(255),
    ->     city VARCHAR(100),
    ->     state VARCHAR(100),
    ->     country VARCHAR(100),
    ->     contact_email VARCHAR(150),
    ->     contact_phone VARCHAR(30),
    ->     established_year YEAR,
    ->
    ->     PRIMARY KEY (museum_id)
    -> );
Query OK, 0 rows affected (0.07 sec)

mysql> CREATE TABLE COLLECTION (
    ->     collection_id INT AUTO_INCREMENT,
    ->     museum_id INT NOT NULL,
    ->     collection_name VARCHAR(150) NOT NULL,
    ->     description TEXT,
    ->     historical_period VARCHAR(100),
    ->     collection_type VARCHAR(100),
    ->     date_created DATE,
    ->
    ->     PRIMARY KEY (collection_id),
    ->
    ->     CONSTRAINT fk_collection_museum
    ->         FOREIGN KEY (museum_id)
    ->         REFERENCES MUSEUM(museum_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> CREATE TABLE CREATOR (
    ->     creator_id INT NOT NULL,
    ->     creator_name VARCHAR(150) NOT NULL,
    ->     birth_year YEAR,
    ->     nationality VARCHAR(100),
    ->     biography TEXT,
    ->
    ->     PRIMARY KEY (creator_id)
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE ARTIFACT (
    ->     artifact_id INT AUTO_INCREMENT,
    ->     collection_id INT NOT NULL,
    ->     creator_id INT,
    ->     accession_number VARCHAR(100) NOT NULL,
    ->     artifact name VARCHAR(200) NOT NULL,
    ->     description TEXT,
    ->     artifact_type VARCHAR(100),
    ->     origin_place VARCHAR(150),
    ->     creation_date DATE,
    ->     estimated_value DECIMAL(15,2),
    ->     acquisition_date DATE,
    ->     status VARCHAR(100),
    ->
    ->     PRIMARY KEY (artifact_id),
    ->
    ->     CONSTRAINT uq_artifact_accession UNIQUE (accession_number),
    ->
    ->     CONSTRAINT fk_artifact_collection FOREIGN KEY (collection_id) REFERENCES COLLECTION(collection_id) ON UPDATE CASCADE ON DELETE RESTRICT,
    ->
    ->     CONSTRAINT fk_artifact_creator FOREIGN KEY (creator_id) REFERENCES CREATOR(creator_id) ON UPDATE CASCADE ON DELETE SET NULL
    -> );
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> CREATE TABLE STAFF (
    ->     staff_id INT AUTO_INCREMENT,
    ->     staff_name VARCHAR(150) NOT NULL,
    ->     email VARCHAR(150) NOT NULL,
    ->     phone VARCHAR(30),
    ->     designation VARCHAR(100),
    ->     department VARCHAR(100),
    ->     username VARCHAR(100) NOT NULL,
    ->     password_hash VARCHAR(255) NOT NULL,
    ->     staff_status VARCHAR(50),
    ->
    ->     PRIMARY KEY (staff_id),
    ->
    ->     CONSTRAINT uq_staff_email
    ->         UNIQUE (email),
    ->
    ->     CONSTRAINT uq_staff_username
    ->         UNIQUE (username)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE EXHIBITION (
    ->     exhibition_id INT NOT NULL,
    ->     museum_id INT NOT NULL,
    ->     exhibition_name VARCHAR(200) NOT NULL,
    ->     description TEXT,
    ->     theme VARCHAR(150),
    ->     start_date DATE,
    ->     end_date DATE,
    ->     status VARCHAR(50),
    ->
    ->     PRIMARY KEY (exhibition_id),
    ->
    ->     CONSTRAINT fk_exhibition_museum
    ->         FOREIGN KEY (museum_id)
    ->         REFERENCES MUSEUM(museum_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> CREATE TABLE DISPLAY (
    ->     exhibition_id INT NOT NULL,
    ->     artifact_id INT NOT NULL,
    ->     display_start_date DATE,
    ->     display_end_date DATE,
    ->     display_location VARCHAR(150),
    ->     display_case VARCHAR(100),
    ->     display_notes TEXT,
    ->
    ->     CONSTRAINT fk_display_exhibition
    ->         FOREIGN KEY (exhibition_id)
    ->         REFERENCES EXHIBITION(exhibition_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE CASCADE,
    ->
    ->     CONSTRAINT fk_display_artifact
    ->         FOREIGN KEY (artifact_id)
    ->         REFERENCES ARTIFACT(artifact_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> CREATE TABLE ARTIFACT_MEDIA (
    ->     media_id INT NOT NULL,
    ->     artifact_id INT NOT NULL,
    ->     staff_id INT,
    ->     media_type VARCHAR(50),
    ->     file_name VARCHAR(255),
    ->     file_url VARCHAR(500),
    ->     description TEXT,
    ->     uploaded_on DATETIME DEFAULT CURRENT_TIMESTAMP,
    ->     file_size BIGINT,
    ->
    ->     PRIMARY KEY (media_id),
    ->
    ->     CONSTRAINT fk_media_artifact
    ->         FOREIGN KEY (artifact_id)
    ->         REFERENCES ARTIFACT(artifact_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE CASCADE,
    ->
    ->     CONSTRAINT fk_media_staff
    ->         FOREIGN KEY (staff_id)
    ->         REFERENCES STAFF(staff_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE SET NULL
    -> );
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> CREATE TABLE ARTIFACT_HISTORY (
    ->     history_id INT AUTO_INCREMENT,
    ->     artifact_id INT NOT NULL,
    ->     staff_id INT,
    ->     event_date DATE NOT NULL,
    ->     event_type VARCHAR(100),
    ->     location VARCHAR(150),
    ->     description TEXT,
    ->     source_reference VARCHAR(500),
    ->
    ->     PRIMARY KEY (history_id),
    ->
    ->     CONSTRAINT fk_history_artifact
    ->         FOREIGN KEY (artifact_id)
    ->         REFERENCES ARTIFACT(artifact_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE CASCADE,
    ->
    ->     CONSTRAINT fk_history_staff
    ->         FOREIGN KEY (staff_id)
    ->         REFERENCES STAFF(staff_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE SET NULL
    -> );
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> CREATE TABLE RESTORATION_PROJECT (
    ->     project_id INT AUTO_INCREMENT,
    ->     artifact_id INT NOT NULL,
    ->     manager_id INT,
    ->     project_name VARCHAR(200) NOT NULL,
    ->     objective TEXT,
    ->     start_date DATE,
    ->     end_date DATE,
    ->     status VARCHAR(50),
    ->     cost DECIMAL(15,2),
    ->
    ->     PRIMARY KEY (project_id),
    ->
    ->     CONSTRAINT fk_project_artifact
    ->         FOREIGN KEY (artifact_id)
    ->         REFERENCES ARTIFACT(artifact_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE CASCADE,
    ->
    ->     CONSTRAINT fk_project_manager
    ->         FOREIGN KEY (manager_id)
    ->         REFERENCES STAFF(staff_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE SET NULL
    -> );
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> CREATE TABLE CONDITION_ASSESSMENT (
    ->     assessment_id INT AUTO_INCREMENT,
    ->     artifact_id INT NOT NULL,
    ->     staff_id INT,
    ->     assessment_date DATE NOT NULL,
    ->     conditions VARCHAR(100),
    ->     damage_type VARCHAR(100),
    ->     damage_description TEXT,
    ->     recommended_action TEXT,
    ->     priority VARCHAR(50),
    ->
    ->     PRIMARY KEY (assessment_id),
    ->
    ->     CONSTRAINT fk_assessment_artifact
    ->         FOREIGN KEY (artifact_id)
    ->         REFERENCES ARTIFACT(artifact_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE CASCADE,
    ->
    ->     CONSTRAINT fk_assessment_staff
    ->         FOREIGN KEY (staff_id)
    ->         REFERENCES STAFF(staff_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE SET NULL
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE RESTORATION_EXPERT (
    ->     expert_id INT AUTO_INCREMENT,
    ->     expert_name VARCHAR(150) NOT NULL,
    ->     email VARCHAR(150),
    ->     phone VARCHAR(30),
    ->     hours_worked DECIMAL(8,2),
    ->
    ->     PRIMARY KEY (expert_id)
    -> );
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> CREATE TABLE PROJECT_EXPERT (
    ->     project_id INT NOT NULL,
    ->     expert_id INT NOT NULL,
    ->
    ->     CONSTRAINT fk_project_expert_project
    ->         FOREIGN KEY (project_id)
    ->         REFERENCES RESTORATION_PROJECT(project_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE CASCADE,
    ->
    ->     CONSTRAINT fk_project_expert_expert
    ->         FOREIGN KEY (expert_id)
    ->         REFERENCES RESTORATION_EXPERT(expert_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE RESTORATION_MATERIAL (
    ->     material_id INT AUTO_INCREMENT,
    ->     material_name VARCHAR(150) NOT NULL,
    ->     material_type VARCHAR(100),
    ->     supplier VARCHAR(150),
    ->     unit VARCHAR(50),
    ->     unit_cost DECIMAL(12,2),
    ->     quantity_used DECIMAL(10,2),
    ->     total_cost DECIMAL(15,2),
    ->
    ->     PRIMARY KEY (material_id)
    -> );
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> CREATE TABLE PROJECT_MATERIAL (
    ->     project_id INT NOT NULL,
    ->     material_id INT NOT NULL,
    ->
    ->     CONSTRAINT fk_project_material_project
    ->         FOREIGN KEY (project_id)
    ->         REFERENCES RESTORATION_PROJECT(project_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE CASCADE,
    ->
    ->     CONSTRAINT fk_project_material_material
    ->         FOREIGN KEY (material_id)
    ->         REFERENCES RESTORATION_MATERIAL(material_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE VISITOR (
    ->     visitor_id INT AUTO_INCREMENT,
    ->     visitor_name VARCHAR(150) NOT NULL,
    ->     email VARCHAR(150),
    ->     phone VARCHAR(30),
    ->     age INT,
    ->     nationality VARCHAR(100),
    ->     registration_date DATE,
    ->
    ->     PRIMARY KEY (visitor_id)
    -> );
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> CREATE TABLE VISIT_RECORD (
    ->     visit_id INT AUTO_INCREMENT,
    ->     visitor_id INT NOT NULL,
    ->     museum_id INT NOT NULL,
    ->     visit_date DATE NOT NULL,
    ->     entry_time TIME,
    ->     exit_time TIME,
    ->     visit_type VARCHAR(50),
    ->     ticket_type VARCHAR(50),
    ->
    ->     PRIMARY KEY (visit_id),
    ->
    ->     CONSTRAINT fk_visit_visitor
    ->         FOREIGN KEY (visitor_id)
    ->         REFERENCES VISITOR(visitor_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE CASCADE,
    ->
    ->     CONSTRAINT fk_visit_museum
    ->         FOREIGN KEY (museum_id)
    ->         REFERENCES MUSEUM(museum_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> show tables;
+-----------------------------+
| Tables_in_digital_museum_db |
+-----------------------------+
| artifact                    |
| artifact_history            |
| artifact_media              |
| collection                  |
| condition_assessment        |
| creator                     |
| display                     |
| exhibition                  |
| museum                      |
| project_expert              |
| project_material            |
| restoration_expert          |
| restoration_material        |
| restoration_project         |
| staff                       |
| visit_record                |
| visitor                     |
+-----------------------------+
17 rows in set (0.02 sec)
```

```
mysql> desc artifact;
+------------------+---------------+------+-----+---------+----------------+
| Field            | Type          | Null | Key | Default | Extra          |
+------------------+---------------+------+-----+---------+----------------+
| artifact_id      | int           | NO   | PRI | NULL    | auto_increment |
| collection_id    | int           | NO   | MUL | NULL    |                |
| creator_id       | int           | YES  | MUL | NULL    |                |
| accession_number | varchar(100)  | NO   | UNI | NULL    |                |
| artifact_name    | varchar(200)  | NO   |     | NULL    |                |
| description      | text          | YES  |     | NULL    |                |
| artifact_type    | varchar(100)  | YES  |     | NULL    |                |
| origin_place     | varchar(150)  | YES  |     | NULL    |                |
| creation_date    | date          | YES  |     | NULL    |                |
| estimated_value  | decimal(15,2) | YES  |     | NULL    |                |
| acquisition_date | date          | YES  |     | NULL    |                |
| status           | varchar(100)  | YES  |     | NULL    |                |
+------------------+---------------+------+-----+---------+----------------+
12 rows in set (0.01 sec)

mysql> desc artifact_history;
+------------------+--------------+------+-----+---------+----------------+
| Field            | Type         | Null | Key | Default | Extra          |
+------------------+--------------+------+-----+---------+----------------+
| history_id       | int          | NO   | PRI | NULL    | auto_increment |
| artifact_id      | int          | NO   | MUL | NULL    |                |
| staff_id         | int          | YES  | MUL | NULL    |                |
| event_date       | date         | NO   |     | NULL    |                |
| event_type       | varchar(100) | YES  |     | NULL    |                |
| location         | varchar(150) | YES  |     | NULL    |                |
| description      | text         | YES  |     | NULL    |                |
| source_reference | varchar(500) | YES  |     | NULL    |                |
+------------------+--------------+------+-----+---------+----------------+
8 rows in set (0.00 sec)
```

```
mysql> desc artifact_media;
+-------------+--------------+------+-----+-------------------+-------------------+
| Field       | Type         | Null | Key | Default           | Extra             |
+-------------+--------------+------+-----+-------------------+-------------------+
| media_id    | int          | NO   | PRI | NULL              |                   |
| artifact_id | int          | NO   | MUL | NULL              |                   |
| staff_id    | int          | YES  | MUL | NULL              |                   |
| media_type  | varchar(50)  | YES  |     | NULL              |                   |
| file_name   | varchar(255) | YES  |     | NULL              |                   |
| file_url    | varchar(500) | YES  |     | NULL              |                   |
| description | text         | YES  |     | NULL              |                   |
| uploaded_on | datetime     | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
| file_size   | bigint       | YES  |     | NULL              |                   |
+-------------+--------------+------+-----+-------------------+-------------------+
9 rows in set (0.01 sec)

mysql> desc collection;
+-------------------+--------------+------+-----+---------+----------------+
| Field             | Type         | Null | Key | Default | Extra          |
+-------------------+--------------+------+-----+---------+----------------+
| collection_id     | int          | NO   | PRI | NULL    | auto_increment |
| museum_id         | int          | NO   | MUL | NULL    |                |
| collection_name   | varchar(150) | NO   |     | NULL    |                |
| description       | text         | YES  |     | NULL    |                |
| historical_period | varchar(100) | YES  |     | NULL    |                |
| collection_type   | varchar(100) | YES  |     | NULL    |                |
| date_created      | date         | YES  |     | NULL    |                |
+-------------------+--------------+------+-----+---------+----------------+
```

```
mysql> desc condition_assessment;
+--------------------+--------------+------+-----+---------+----------------+
| Field              | Type         | Null | Key | Default | Extra          |
+--------------------+--------------+------+-----+---------+----------------+
| assessment_id      | int          | NO   | PRI | NULL    | auto_increment |
| artifact_id        | int          | NO   | MUL | NULL    |                |
| staff_id           | int          | YES  | MUL | NULL    |                |
| assessment_date    | date         | NO   |     | NULL    |                |
| conditions         | varchar(100) | YES  |     | NULL    |                |
| damage_type        | varchar(100) | YES  |     | NULL    |                |
| damage_description | text         | YES  |     | NULL    |                |
| recommended_action | text         | YES  |     | NULL    |                |
| priority           | varchar(50)  | YES  |     | NULL    |                |
+--------------------+--------------+------+-----+---------+----------------+
9 rows in set (0.00 sec)

mysql> desc crator;
ERROR 1146 (42S02): Table 'digital_museum_db.crator' doesn't exist
mysql> desc creator;
+--------------+--------------+------+-----+---------+-------+
| Field        | Type         | Null | Key | Default | Extra |
+--------------+--------------+------+-----+---------+-------+
| creator_id   | int          | NO   | PRI | NULL    |       |
| creator_name | varchar(150) | NO   |     | NULL    |       |
| birth_year   | year         | YES  |     | NULL    |       |
| nationality  | varchar(100) | YES  |     | NULL    |       |
| biography    | text         | YES  |     | NULL    |       |
+--------------+--------------+------+-----+---------+-------+
5 rows in set (0.00 sec)
```

```
mysql> desc display;
+--------------------+--------------+------+-----+---------+-------+
| Field              | Type         | Null | Key | Default | Extra |
+--------------------+--------------+------+-----+---------+-------+
| exhibition_id      | int          | NO   | MUL | NULL    |       |
| artifact_id        | int          | NO   | MUL | NULL    |       |
| display_start_date | date         | YES  |     | NULL    |       |
| display_end_date   | date         | YES  |     | NULL    |       |
| display_location   | varchar(150) | YES  |     | NULL    |       |
| display_case       | varchar(100) | YES  |     | NULL    |       |
| display_notes      | text         | YES  |     | NULL    |       |
+--------------------+--------------+------+-----+---------+-------+
7 rows in set (0.01 sec)

mysql> desc exhibition;
+-----------------+--------------+------+-----+---------+-------+
| Field           | Type         | Null | Key | Default | Extra |
+-----------------+--------------+------+-----+---------+-------+
| exhibition_id   | int          | NO   | PRI | NULL    |       |
| museum_id       | int          | NO   | MUL | NULL    |       |
| exhibition_name | varchar(200) | NO   |     | NULL    |       |
| description     | text         | YES  |     | NULL    |       |
| theme           | varchar(150) | YES  |     | NULL    |       |
| start_date      | date         | YES  |     | NULL    |       |
| end_date        | date         | YES  |     | NULL    |       |
| status          | varchar(50)  | YES  |     | NULL    |       |
+-----------------+--------------+------+-----+---------+-------+
8 rows in set (0.00 sec)
```

```
mysql> desc museum;
+------------------+--------------+------+-----+---------+----------------+
| Field            | Type         | Null | Key | Default | Extra          |
+------------------+--------------+------+-----+---------+----------------+
| museum_id        | int          | NO   | PRI | NULL    | auto_increment |
| museum_name      | varchar(150) | NO   |     | NULL    |                |
| description      | text         | YES  |     | NULL    |                |
| address          | varchar(255) | YES  |     | NULL    |                |
| city             | varchar(100) | YES  |     | NULL    |                |
| state            | varchar(100) | YES  |     | NULL    |                |
| country          | varchar(100) | YES  |     | NULL    |                |
| contact_email    | varchar(150) | YES  |     | NULL    |                |
| contact_phone    | varchar(30)  | YES  |     | NULL    |                |
| established_year | year         | YES  |     | NULL    |                |
+------------------+--------------+------+-----+---------+----------------+
10 rows in set (0.01 sec)

mysql> desc project_expert;
+------------+------+------+-----+---------+-------+
| Field      | Type | Null | Key | Default | Extra |
+------------+------+------+-----+---------+-------+
| project_id | int  | NO   | MUL | NULL    |       |
| expert_id  | int  | NO   | MUL | NULL    |       |
+------------+------+------+-----+---------+-------+
2 rows in set (0.00 sec)

mysql> desc project_material;
+-------------+------+------+-----+---------+-------+
| Field       | Type | Null | Key | Default | Extra |
+-------------+------+------+-----+---------+-------+
| project_id  | int  | NO   | MUL | NULL    |       |
| material_id | int  | NO   | MUL | NULL    |       |
+-------------+------+------+-----+---------+-------+
2 rows in set (0.01 sec)
```

```
mysql> desc restoration_expert;
+--------------+--------------+------+-----+---------+----------------+
| Field        | Type         | Null | Key | Default | Extra          |
+--------------+--------------+------+-----+---------+----------------+
| expert_id    | int          | NO   | PRI | NULL    | auto_increment |
| expert_name  | varchar(150) | NO   |     | NULL    |                |
| email        | varchar(150) | YES  |     | NULL    |                |
| phone        | varchar(30)  | YES  |     | NULL    |                |
| hours_worked | decimal(8,2) | YES  |     | NULL    |                |
+--------------+--------------+------+-----+---------+----------------+
5 rows in set (0.01 sec)

mysql> desc restoration_material;
+---------------+---------------+------+-----+---------+----------------+
| Field         | Type          | Null | Key | Default | Extra          |
+---------------+---------------+------+-----+---------+----------------+
| material_id   | int           | NO   | PRI | NULL    | auto_increment |
| material_name | varchar(150)  | NO   |     | NULL    |                |
| material_type | varchar(100)  | YES  |     | NULL    |                |
| supplier      | varchar(150)  | YES  |     | NULL    |                |
| unit          | varchar(50)   | YES  |     | NULL    |                |
| unit_cost     | decimal(12,2) | YES  |     | NULL    |                |
| quantity_used | decimal(10,2) | YES  |     | NULL    |                |
| total_cost    | decimal(15,2) | YES  |     | NULL    |                |
+---------------+---------------+------+-----+---------+----------------+
8 rows in set (0.01 sec)
```

```
mysql> desc restoration_project;
+--------------+---------------+------+-----+---------+----------------+
| Field        | Type          | Null | Key | Default | Extra          |
+--------------+---------------+------+-----+---------+----------------+
| project_id   | int           | NO   | PRI | NULL    | auto_increment |
| artifact_id  | int           | NO   | MUL | NULL    |                |
| manager_id   | int           | YES  | MUL | NULL    |                |
| project_name | varchar(200)  | NO   |     | NULL    |                |
| objective    | text          | YES  |     | NULL    |                |
| start_date   | date          | YES  |     | NULL    |                |
| end_date     | date          | YES  |     | NULL    |                |
| status       | varchar(50)   | YES  |     | NULL    |                |
| cost         | decimal(15,2) | YES  |     | NULL    |                |
+--------------+---------------+------+-----+---------+----------------+
9 rows in set (0.00 sec)

mysql> desc staff;
+---------------+--------------+------+-----+---------+----------------+
| Field         | Type         | Null | Key | Default | Extra          |
+---------------+--------------+------+-----+---------+----------------+
| staff_id      | int          | NO   | PRI | NULL    | auto_increment |
| staff_name    | varchar(150) | NO   |     | NULL    |                |
| email         | varchar(150) | NO   | UNI | NULL    |                |
| phone         | varchar(30)  | YES  |     | NULL    |                |
| designation   | varchar(100) | YES  |     | NULL    |                |
| department    | varchar(100) | YES  |     | NULL    |                |
| username      | varchar(100) | NO   | UNI | NULL    |                |
| password_hash | varchar(255) | NO   |     | NULL    |                |
| staff_status  | varchar(50)  | YES  |     | NULL    |                |
+---------------+--------------+------+-----+---------+----------------+
9 rows in set (0.00 sec)
```

```
mysql> desc visitor;
+-------------------+--------------+------+-----+---------+----------------+
| Field             | Type         | Null | Key | Default | Extra          |
+-------------------+--------------+------+-----+---------+----------------+
| visitor_id        | int          | NO   | PRI | NULL    | auto_increment |
| visitor_name      | varchar(150) | NO   |     | NULL    |                |
| email             | varchar(150) | YES  |     | NULL    |                |
| phone             | varchar(30)  | YES  |     | NULL    |                |
| age               | int          | YES  |     | NULL    |                |
| nationality       | varchar(100) | YES  |     | NULL    |                |
| registration_date | date         | YES  |     | NULL    |                |
+-------------------+--------------+------+-----+---------+----------------+
7 rows in set (0.00 sec)

mysql> desc visit_record;
+-------------+-------------+------+-----+---------+----------------+
| Field       | Type        | Null | Key | Default | Extra          |
+-------------+-------------+------+-----+---------+----------------+
| visit_id    | int         | NO   | PRI | NULL    | auto_increment |
| visitor_id  | int         | NO   | MUL | NULL    |                |
| museum_id   | int         | NO   | MUL | NULL    |                |
| visit_date  | date        | NO   |     | NULL    |                |
| entry_time  | time        | YES  |     | NULL    |                |
| exit_time   | time        | YES  |     | NULL    |                |
| visit_type  | varchar(50) | YES  |     | NULL    |                |
| ticket_type | varchar(50) | YES  |     | NULL    |                |
+-------------+-------------+------+-----+---------+----------------+
8 rows in set (0.00 sec)
```